

Tau Pathology in Alzheimer's Disease Uniquely Affects Sulcal Depths

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Objective: Though it is widely known that tau deposition affects brain structure, the precise localization of these effects is poorly understood, especially in relation to gyral and sulcal anatomy. We investigated whether tau pathology in Alzheimer's disease (AD) preferentially affects sulci, and particularly sulcal depths.

Methods: We analyzed 675 participants from the Alzheimer's Disease Neuroimaging Initiative (ADNI) with magnetic resonance imaging (MRI) and positron emission tomography (PET) data to investigate relationships between neocortical tau PET signal and cortical thickness. We then examined an advanced AD case with postmortem MRI and coregistered whole-brain phospho-tau staining for evidence of sulcal tau distribution in AD. Finally, in a sample of 187 cognitively unimpaired young and older adults with resting-state functional MRI, we examined connectivity strength between tau-vulnerable regions and the hippocampus across adulthood, prior to disease-related cognitive decline.

Results: Our findings revealed that tau-related cortical thinning predominantly occurs in sulcal regions, especially the deepest parts. Postmortem histology confirmed preferential tau accumulation in sulcal depths. Additionally, connectivity analyses revealed that, across adulthood, these primarily sulcal regions most susceptible to tau-related thinning also have stronger connectivity to the hippocampus, suggesting a role for network connectivity in the vulnerability of sulci to the effects of tau pathology later in life.

Interpretation: These findings support the hypothesis that sulci, and particularly their depths, represent structurally and functionally vulnerable regions for tau deposition in AD. Understanding the mechanisms underlying this sulcal vulnerability provides insight into general principles driving regional susceptibility to pathology, and sheds light on the detrimental functional and cognitive effects of tau pathology.

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Alzheimer's disease (AD) is pathologically characterized by the aggregation of misfolded proteins, namely the generalized accumulation of amyloid-beta ($A\beta$) plaques and the stereotyped spread of pathological hyperphosphorylated tau.^{1,2} Though $A\beta$ pathology is associated with degeneration and cognitive decline, the relationship among tau, atrophy, and cognitive impairment is substantially stronger.^{3,4} However, factors underlying the pattern of regional vulnerability to tau pathology, how $A\beta$ might interact with tau given its spatially distinct aggregation, and the effects of these interactions on cognition are incompletely understood.

The cerebral cortex is convoluted into gyri (raised, convex ridges) separated by sulci (depressed, concave grooves). Topographically, the cortical ribbon can be subdivided into the gyral crown (surface-exposed region), walls (sloping sides descending into sulci), and fundus (the bottom of the sulcus).⁵ This nomenclature is not always consistent across studies; in this work, we use the term "sulci" to refer to the concave, buried portion of the cortical ribbon. Structurally, sulci, particularly in associative regions, have been found to serve as unique markers of morphological and pathological changes associated with both aging and AD, as well as other disorders. Recent work has demonstrated that cortical thinning is greater in sulci than gyri in aging⁶ and that various features of sulcal morphology are associated with cognitive decline.^{7,8} In AD, features of sulcal morphology (such as sulcal width and thickness) are associated with cognitive impairment and, in some cases, may be more diagnostically accurate than traditional structural measures, such as hippocampal volume.^{9,10} In addition, in post-mortem AD samples, $A\beta$ preferentially deposits in sulci.^{11,12} Furthermore, although historically a sulcal tau bias has not been considered a characteristic of AD, a pathological hallmark of chronic traumatic encephalopathy (CTE) is perivascular tau pathology preferentially found in the deepest areas of the sulcal basin (sulcal depths)¹³; in cases with mixed CTE-AD pathologies, both tau and $A\beta$ have been found to aggregate in sulcal depths.¹⁴

Despite these observations, it remains unclear whether tau pathology in AD preferentially affects sulcal regions, as has been observed for $A\beta$ in AD and tau in CTE. Additionally, why sulci appear to be uniquely vulnerable to both age- and AD-related changes, as well as potential effects of this vulnerability, have not been elucidated. One avenue to explore this question stems from the observation that sulci and gyri may have different connectivity patterns: though results are somewhat inconsistent across studies, sulcal depths have been hypothesized to be long-range connectivity hubs¹⁵ and to be cognitively relevant.¹⁶ Differences in the connectivity patterns between sulci and gyri may impact how pathological

proteins spread, with long-range connectivity hubs potentially facilitating the neocortical propagation of tau pathology. Beyond their structural and functional significance, sulci have distinct microstructural, cytoarchitectonic, and mechanical properties¹⁷ that may contribute to their vulnerability in neurodegeneration. Understanding these relationships may help clarify fundamental mechanisms of AD progression and inform future therapeutic strategies targeting structural and functional vulnerabilities in the aging brain. We therefore performed this study to determine whether, in AD, tau pathology and tau-related atrophy tend to affect sulci (and sulcal depths), and whether these relationships might be associated with functional network connectivity patterns.

Materials and Methods

Participants

Written informed consent was obtained for all participants under protocols approved by the institutional review boards of all participating institutions for each study (University of California – San Francisco [UCSF] for postmortem data, UC Berkeley and Lawrence Berkeley National Laboratory for Berkeley Aging Cohort Study [BACS]).

Alzheimer's Disease Neuroimaging Initiative Sample. Data from the section of this study investigating relationships between tau positron emission tomography (PET) and magnetic resonance imaging (MRI) measures are from Alzheimer's Disease Neuroimaging Initiative Sample (ADNI; <https://adni.loni.usc.edu/>); the primary goal of ADNI is to test whether serial MRI, PET, other biological markers, and clinical and neuropsychological assessment can be combined to measure the progression of AD. The ADNI sample in this study consists of the 675 older adults from the following groups who (as of 2023) had a 3T MRI scan, tau ([18F]florbetapir [FTP]) PET scan, and amyloid ([18F]florbetaben [FBB] or [18F]florbetapir [FBP]) PET scan within at most 1 year of each other: 94 subjects with AD (all $A\beta$ +), 113 subjects with $A\beta$ + MCI, and 468 subjects cognitively unimpaired (166 $A\beta$ +). Participant demographics are indicated in Supplementary Table S1. All 675 subjects had memory composite scores available within a year of their MRI. A subset of 91 of the participants included in this analysis had a second T1 scan at the same session date as the one used in the main analyses. To confirm there was no difference between sulci and gyri in measurement noise, we processed these repeated scans and extracted the average thickness of whole-cortex sulcal and gyral regions of interest (ROIs), as well as of each standard FreeSurfer (FS) Desikan-Killiany (DK) atlas region (averaged across both hemispheres).

Postmortem Histology Sample. For histology analyses, we used a specimen described in a prior publication¹⁸ from a 76-year-old donor with a diagnosis of dementia and severe AD pathology at autopsy (Braak tau pathology stage 6, AD neuropathologic changes score A3B3C3).

Berkeley Aging Cohort Study Sample. Resting state functional MRI (rs-fMRI) analyses were performed with data from BACS, an ongoing longitudinal study of cognitive aging in community-dwelling cognitively normal older adults and young adult controls. A 3T structural MRI and 3T rs-fMRI were available for 122 cognitively unimpaired older adults (OAs), aged 60 to 95 years (53 amyloid positive [OA+] and 69 amyloid negative [OA-]) and 65 younger adults (YAs; aged 18–50 years). Participant demographics are indicated in Supplementary Table S2.

Image Acquisition and Processing

Alzheimer's Disease Neuroimaging Initiative: Magnetic Resonance Imaging. In ADNI, anatomic 3T T1-weighted whole-brain magnetization prepared rapid gradient echo (MPRAGE) scans were acquired using a standardized protocol across sites. Cortical surface reconstructions were generated for each participant's MRI using the standard FS processing pipeline (FreeSurfer, version 7.1.0: surfer.nmr.mgh.harvard.edu).^{19,20} Estimated total intracranial volume (eTIV) was determined using the FS estimate. The cortical segmentation was visually inspected for segmentation errors (Supplementary Fig S1).

Alzheimer's Disease Neuroimaging Initiative: Positron Emission Tomography. Amyloid PET scans were smoothed to a resolution of 6 mm and used to define amyloid positivity, which was determined by the ADNI-derived cortical summary region (FS-defined frontal, anterior/posterior cingulate, lateral parietal, and lateral temporal regions) standardized value uptake ratios (SUVRs) with a tracer-specific cutoff, as described in the ADNI documentation. Tau PET scans were smoothed to a resolution of 6 mm, and SUVRs were calculated voxelwise and in FS-defined ROIs. For neocortical tau analyses, FTP SUVR was calculated for a composite neocortical ROI consisting of the following FS DK atlas-defined regions: entorhinal, parahippocampal, fusiform, lingual, amygdala, middle and inferior temporal, caudal and rostral anterior cingulate, posterior and isthmus cingulate, insula, temporal pole, superior frontal, lateral and medial orbitofrontal, frontal pole, rostral and caudal middle frontal, pars opercularis, pars orbitalis, pars triangularis, lateral occipital, supramarginal and inferior parietal, superior temporal, superior parietal, precuneus, banks of the superior temporal sulcus, transverse temporal, pericalcarine, posterior central, cuneus, precentral and paracentral.

Sulcal and Gyrus ROIs. Sulcal and gyrus ROIs were generated by thresholding the FS curvature (.curv) file at 0, and were used to mask the FS cortical parcellation into sulcal and gyrus subsets. The sulcal ROIs were further split into sulcal depths versus the rest of the sulcus, determined by thresholding the deepest 5% of vertices versus the remaining 95% of vertices for each subject, based on FS depth (.sulc) maps.

Postmortem Histology and MRI. The postmortem brain specimen underwent structural spoiled gradient echo (SPGR) T1-weighted 3T MRI acquisition within 10 hours of death,

followed by autopsy, embedding and coronal sectioning, and immunohistochemical staining for AT8, a monoclonal antibody against phospho-Tau (Ser202, Thr205); the full image processing pipeline is detailed in Ushizima et al, 2022¹⁸ (see Supplementary Fig S2). The previously described AT8 heatmap ($1 \times 1 \mu\text{m}^2$ squares spanning every four 160 μm slices) registered to the postmortem MRI processed through FS were used for analyses. Average AT8 heatmap intensity was calculated for every DK atlas region masked into sulcal/gyrus components, with the exception of the following regions, which were excluded following manual inspection of both the histological sections and MRI surface reconstructions (due to a lesion or issues with the staining and/or reconstruction): left-hemisphere banks of the superior temporal sulcus, fusiform, inferior parietal/temporal, pericalcarine; left- and right-hemisphere cuneus, lateral occipital, and lingual.

BACS and Resting State Functional Connectivity. Structural and functional MRI data were collected on a 3T Siemens TIM/Trio scanner (software version B17A; Siemens, Erlangen, Germany), as described previously.²¹ T1-weighted MPRAGE anatomic scans (1 mm isotropic voxels; TR = 2,300 ms, TE = 2.98 ms) were collected for each participant and processed with FS (version 7.1.0). The rs-fMRI scans were obtained using T2*-weighted echo planar imaging (EPI; 2.6 mm isotropic voxels, TR = 1.067 s, TE = 31.2 ms). FC strength was calculated using the CONN functional connectivity toolbox (version 17e)²² in MATLAB (version 2021a), with preprocessing described previously.²¹ In brief, motion detection was performed using ART (https://www.nitrc.org/projects/artifact_detect), with a movement threshold of 0.5 mm/TR and a global intensity z-score threshold of 3. Volumes identified as outliers were flagged and used as spike regressors in denoising, which was performed using translation and rotation realignment parameters and their first-order derivatives, along with anatomic CompCor. A band pass filter of 0.008 to 0.1 hertz (Hz) and linear detrending were then applied to the residual time-series. Within each individual, the mean denoised time series was extracted from a DK-atlas-defined hippocampal seed ROI and correlated (Pearson's r) with the time series of each cortical voxel. The resulting hippocampal seed-to-voxel connectivity maps were used to compute average connectivity to the hippocampus in subsequent analyses. We chose a hippocampal seed because prior work has shown that functional connectivity between the hippocampus and key regions of downstream tau spread is associated with tau accumulation in those regions.^{21,23}

Statistical Analysis

Tau-Thickness General Linear Models: Neocortical Tau Models. Correlation between vertex-wise cortical thickness (with thickness maps smoothed with a 3 mm full-width/half-max Gaussian kernel) and neocortical tau SUVR were performed using the FS General Linear Model (GLM) pipeline, generating an ROI of significant clusters. Cluster-wise multiple comparisons correction was performed via simulation with *mri_glmfit-sim*, with a cluster-wise p value threshold of 0.05 and a vertex-wise

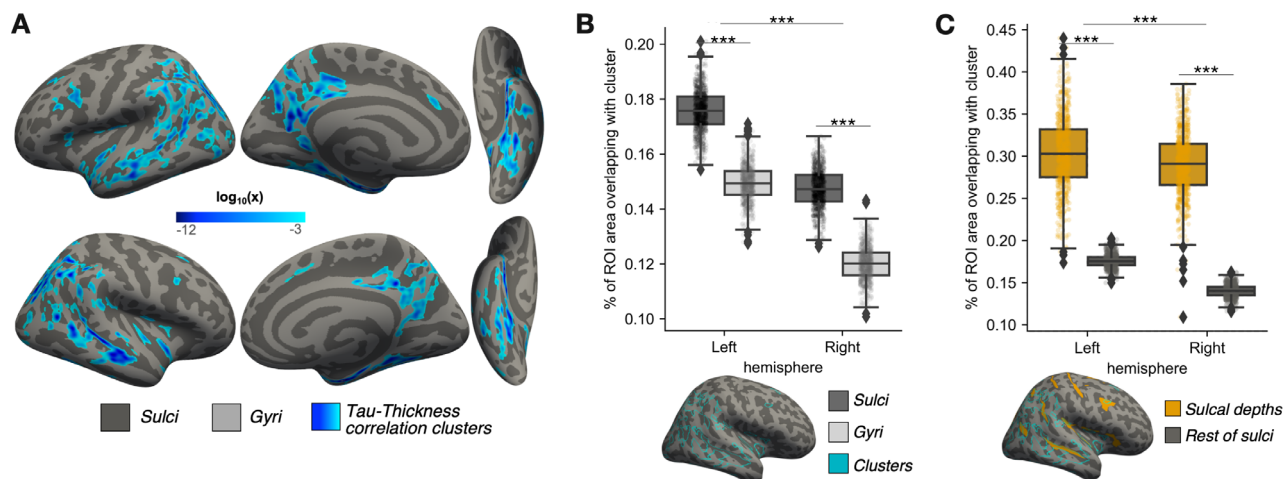


FIGURE 1: Tau-thickness correlation clusters are mostly sulcal. (A) Clusters of significant vertex-wise cortical thickness and neocortical tau (FTP) SUVR correlations, with significance values (uncorrected) masked by clusters surviving multiple comparison correction (vertex-wise cluster threshold = 0.001, cluster-wise p value threshold = 0.05). (B) Percent of area of sulcal or gyral ROIs overlapping with binarized clusters from A. Mixed-effects ANOVA shows a main effect of ROI ($p < 0.0001$), hemisphere ($p < 0.0001$), and an ROI $\times$ hemisphere interaction ($p < 0.0001$). Bottom: Binarized clusters from A projected to an example individual surface. (C) Percent of area of sulcal depths (deepest 5%) or the remaining 95% of sulcal points overlapping with clusters from A. Mixed-effects ANOVA shows a main effect of ROI ($p < 0.0001$), hemisphere ($p < 0.0001$), and an ROI $\times$ hemisphere interaction ($p < 0.001$). Bottom: Clusters binarized and projected to an example individual surface with sulcal depths also shown. *** = $p < 0.0001$. ANOVA = analysis of variance; FTP = flortaucipir; ROI = region of interest; SUVR = standardized value uptake ratio.

cluster forming threshold (CFT) of 0.001, chosen as the maximum values that still minimize false positive rates.²⁴ These tau-thickness clusters were binarized to serve as an ROI in subsequent analyses ("clusters;" Fig 1A) and then projected into each individual subject's space, where the overlap between the clusters and the following ROIs was calculated: whole-brain sulci versus gyri (determined by thresholding the FS .curv file at 0), sulcal depths versus the rest of the sulci (determined by thresholding each sulcal ROI by the deepest 5% of vertices using the FS .sulc file, or the remaining 95% of vertices). Mixed-effects analysis of variances (ANOVAs) were used to examine effects of ROI (sulci/gyri, depths/rest) and hemisphere on the amount of overlap with the tau-thickness clusters. Because the choice of CFT is somewhat subjective, additional CFTs were used to examine the extent of these correlations at different thresholds (Supplementary Fig S3A); larger thresholds than our chosen 0.001 are not recommended because of an inflated risk of false positive results,²⁴ but we also repeated analyses with a more stringent threshold of 0.0001 (Supplementary Fig S4).

Tau-Thickness General Linear Models: Vertex-Wise Tau Models. A GLM comparing tau PET SUVR and thickness (smoothed to 5 mm) was fitted at each vertex in fsaverage6 space to examine relationships between tau PET and cortical thickness at a vertex-wise level. FS-modified Müller-Gärtner partial volume correction (PVC) was applied to the tau PET images. The p values were false discovery rate (FDR) corrected, and also corrected for cluster-wise multiple comparisons with FSL's Permutation Analysis of Linear Models (PALM; https://web.mit.edu/fsl_v5.0.10/fsl/doc/wiki/PALM.html).

Volume-Memory Regression Models. We used multiple linear regression to model ADNI-MEM memory composite scores as a function of ROI volume (of tau-thickness clusters or the rest of the brain) in all ADNI participants with the closest available composite score within 1 year of their MRI ($n = 675$). Volumes were z-scored prior to modeling. First, 2 separate regression models were constructed: one predicting memory scores from the volume of the tau-thickness cluster ROI, and one from the volume of the rest of the cortex, each also controlling for eTIV, age at scan, sex, and years of education. An additional combined model included ROI volume, ROI type (cluster or rest), and their interaction, as well as the aforementioned covariates, to test whether the relationship between volume and memory differed by ROI type. Adjusted relationships were visualized by residualizing both predictors and outcome against covariates and plotting the residuals.

Postmortem Histology. We used t tests to compare the average heatmap signal in sulcal and gyral ROIs, and in sulcal depths versus the rest of sulci.

rsFC Analyses. The cluster ROI from the neocortical tau-thickness GLM in ADNI was projected into each participant's native space volume using FS and used to mask the hippocampal seed-to-voxel FC maps; average FC was calculated by averaging the maps within the cluster ROI or the rest of the cortex. Mixed-effects ANOVAs were used to examine effects of ROI (clusters/rest of cortex) and group (YA, OA-, and OA+) on the hippocampal FC strength.

Packages Used for Statistical Analysis. All statistical tests and visualization were implemented in Python (version 3.9.13) and R (version 4.0.1). For the tau-thickness overlap models, LMEs were implemented with the “lme” function from the “nlme” R software package. ANOVA F tests were implemented with the “anova” function from the built-in stats R package, and the effect sizes for ANOVA effects are reported as partial eta-squared (η^2_p) values computed with the “eta_squared” function from the “effectsize” R software package. Post hoc analyses on ANOVA effects were computed with the “emmeans” and contrast functions from the “emmeans” R software package (with p values adjusted with Tukey’s method). The volume-memory regression models were implemented with the “lm” function from base R software. Two-sided t tests were performed in Python with the “ttest_ind” function from SciPy.stats. The intraclass correlation coefficient (ICC[2, 1]) was calculated using the “intraclass_corr” function from the “pingouin” Python software package. Visualization was done using ggplot2, matplotlib, and Freeview.

Results

Tau-Related Cortical Thinning Is Concentrated in Sulcal Depths

In order to investigate tau-related atrophy patterns, we examined 3T MRI, amyloid, and tau PET data from 675 ADNI participants: 94 with A β -positive (A β +) AD dementia, 113 with A β + mild cognitive impairment (MCI), and 468 cognitively unimpaired (166 A β +) (see Supplementary Table S1). We first implemented a GLM to investigate the correlation between vertex-wise cortical thickness and neocortical tau SUVR across this sample, generating a ROI of significant clusters (“clusters”). These clusters of significant negative tau-thickness correlation generally appeared to recapitulate classic “AD signature” regions of atrophy (Fig 1A and Supplementary Fig S3A).²⁵

To determine whether these tau-thickness clusters are mostly in sulcal areas, we binarized and projected these clusters onto each individual surface and calculated the overlap between the clusters and a whole-cortex sulcal or whole-cortex gyral ROI (see Supplementary Fig S3B), calculated as the percentage of the ROI area that is covered by the clusters (Fig 1B). A linear mixed-effects ANOVA was performed to examine the effects of ROI (sulci vs gyri) and hemisphere (left vs right) on the amount of overlap. ROI, hemisphere, and their interaction were considered fixed effects, with a random intercept for each subject nested within-hemisphere. We observed significant main effects of ROI [$F(1, 1,348) = 12228.6$, $p < 0.0001$, $\eta^2_p = 0.90$] and hemisphere [$F(1, 674) = 14,139.9$, $p < 0.0001$, $\eta^2_p = 0.95$], demonstrating that clusters had greater overlap with sulci than gyri and that the left hemisphere showed greater overlap than the right.

There was also a significant interaction between ROI and hemisphere [$F(1, 1,348) = 6.3$, $p = 0.0121$, $\eta^2_p = 0.0047$; see Fig 1B], although the effect size was small. Post hoc pairwise comparisons revealed that for both sulci and gyri, the left hemisphere overlap was significantly greater than the right (sulci: $p < 0.0001$; gyri: $p < 0.0001$), and that, within each hemisphere, sulci showed greater overlap than gyri (left: $p < 0.0001$; right: $p < 0.0001$). The small but significant interaction suggests that the difference between sulci and gyri is slightly more pronounced in the right hemisphere.

To investigate whether these clusters also mostly tend to be in the sulcal depths, we calculated the amount of overlap with the deepest 5% or the remaining 95% of sulcal points (Fig 1C). An analogous mixed-effects ANOVA examining the effects of ROI (sulcal depths vs rest) and hemisphere on the amount of overlap again revealed significant main effects of ROI [$F(1, 1,348) = 17,646.81$, $p < 0.0001$, $\eta^2_p = 0.93$] and hemisphere [$F(1, 674) = 539.48$, $p < 0.0001$, $\eta^2_p = 0.44$], as well as a significant ROI by hemisphere interaction [$F(1, 1,348) = 115.67$, $p < 0.0001$, $\eta^2_p = 0.08$]. Post hoc comparisons revealed that overlap with the depths was significantly greater ($p < 0.0001$), and also with the left hemisphere ($p < 0.0001$). The significant but small interaction suggests that the difference between ROIs was larger in the right hemisphere compared to the left.

To confirm no difference between sulci and gyri in measurement noise that might affect these findings, we examined a subset of 91 of the subjects who had a repeated MRI scan on the same examination date; an ANOVA comparing whole-cortex sulcal and gyral thickness between scans confirmed no significant difference between the thickness of the sulcal or gyral regions between scans (effect of ROI $\times$ scan $p > 0.1$). Additionally, to examine whether thickness measurements in more gyral regions are less reliable, the ICC of thickness between repeat scans for each DK atlas region was calculated and compared to how sulcal the region was (% of ROI area that is sulcal); there was no significant correlation between how sulcal a region was and its thickness ICC(2, 1) (Pearson $r^2 = 0.068$, $p > 0.1$).

We then extended this examination to the correlation between vertex-wise cortical thickness and vertex-wise tau SUVRs to account for the spatial distribution of tau pathology as well (Supplementary Fig S5). Though clusters of significant correlation are much more widespread at the vertex-wise level, they appear quite similar to the neocortical tau-thickness correlation patterns (see Supplementary Fig S3A), likely due to the well-established strong spatial association between tau and atrophy. We compared the t -statistic in sulcal and gyral areas and found that

sulcal t -values were significantly higher than gyral ($p < 0.001$; see Supplementary Fig S5), suggesting that the tau–thickness association is stronger in sulci. Additionally, to run an analogous comparison in the main analysis model (examining the correlation between vertex-wise cortical thickness and neocortical tau SUVR), we compared the Pearson correlation coefficient (PCC) overlay values from this model in sulci and gyri; the sulcal association strength was significantly more negative compared to gyri (pairwise comparisons $p < 0.001$, Tukey's).

The Volume of Tau-Vulnerable Regions is Associated With Memory

To determine if these tau-vulnerable regions are associated with cognition, we examined whether the volume of the clusters predicts memory (ADNI-MEM memory composite scores) better than the rest of the cortex, while controlling for $eTIV$, age, sex, and education (Fig 2). We used multiple linear regression to assess the association between z-scored ROI volumes and ADNI-MEM scores. First, we ran separate regression models predicting memory performance from ROI volume for each region, including covariates. There was a significant positive relationship between cluster ROI volume and memory ($\beta = 0.430$, adjusted $R^2 = 0.30$, $p < 0.001$), as well as a weaker but still significant relationship for the rest of the cortex ($\beta = 0.255$, adjusted $R^2 = 0.21$, $p < 0.001$). We then ran a combined model to compare the slopes between the ROIs, which showed a significant interaction between ROI type and ROI volume ($\beta = -0.117$, $p = 0.009$), indicating that the slope for the

clusters was significantly greater than the slope for the rest of the cortex. We also compared the relationship between whole-brain sulcal versus gyral volumes and memory, and found that neither predicted memory performance more strongly than the other (interaction $p > 0.2$). These results suggest that volume changes in the primarily sulcal clusters are more strongly associated with memory performance than in the rest of the cortex, highlighting the importance of these regions in cognitive decline.

Postmortem Data Provide Evidence for Sulcal Tau Accumulation

In order to investigate whether this vulnerability of sulcal regions to tau-related thinning occurs because tau pathology preferentially accumulates in sulci in AD, we examined a postmortem Braak 6 case with a 3T postmortem MRI and histological staining with an AT8 phospho-tau antibody (Fig 3A). After co-registering the AT8 heatmap to the MRI, we calculated the average AT8 signal intensity in DK atlas regions masked by a sulcal or gyral mask (Fig 3B); analogous to previous sulcal depth analyses, we also masked sulcal regions by an ROI containing the 5% of deepest sulcal points or the remaining 95% of sulcal area. We found that the AT8 signal in the postmortem case was higher in sulci than gyri ($p < 0.001$), and was also higher in sulcal depths compared to the rest of the sulci ($p < 0.001$; Fig 3C, D).

Primarily Sulcal Tau-Vulnerable Regions Are Highly Connected to the Hippocampus

Last, to examine whether the tau-thickness clusters from the first part of the study (see Fig 1A) represent an early

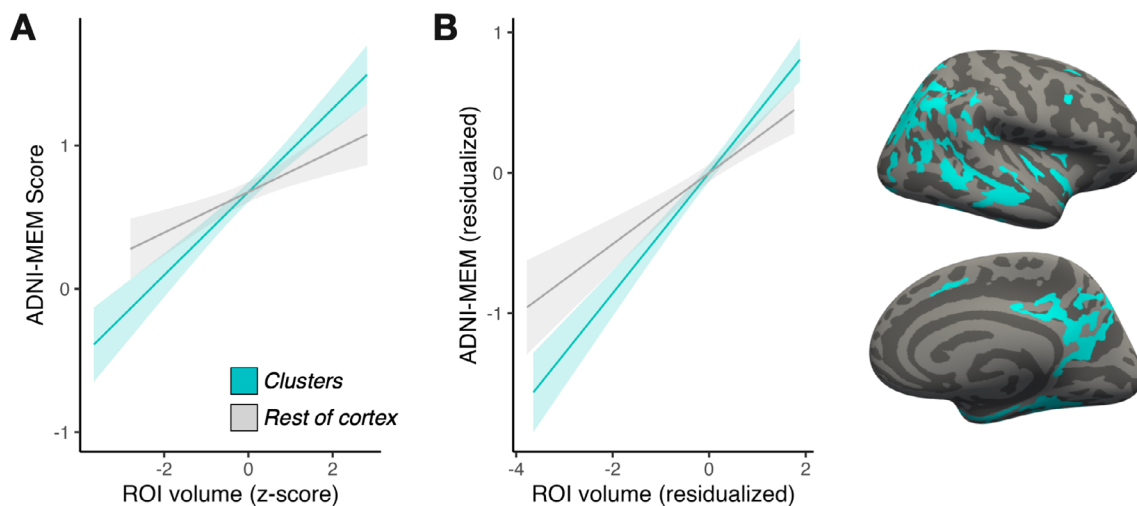


FIGURE 2: Volume of tau-thickness correlation clusters is more strongly related to memory than the rest of the brain. (A) Unadjusted relationships between z-scored ROI volumes (of “Clusters” ROI [binarized tau-thickness correlation clusters] from Fig 1A, or the rest of the cortex) and ADNI-MEM memory composite scores. (B) Residualized plot showing the association between ROI volume and memory after adjusting for age, sex, education, and intracranial volume. In both cases, cluster volume predicts memory scores better than the rest of the cortex: the association is stronger for the clusters (teal) compared to the rest of the cortex (gray), as indicated by both the slopes and the interaction term in a combined model. ADNI = Alzheimer’s Disease Neuroimaging Initiative; ROI = region of interest.

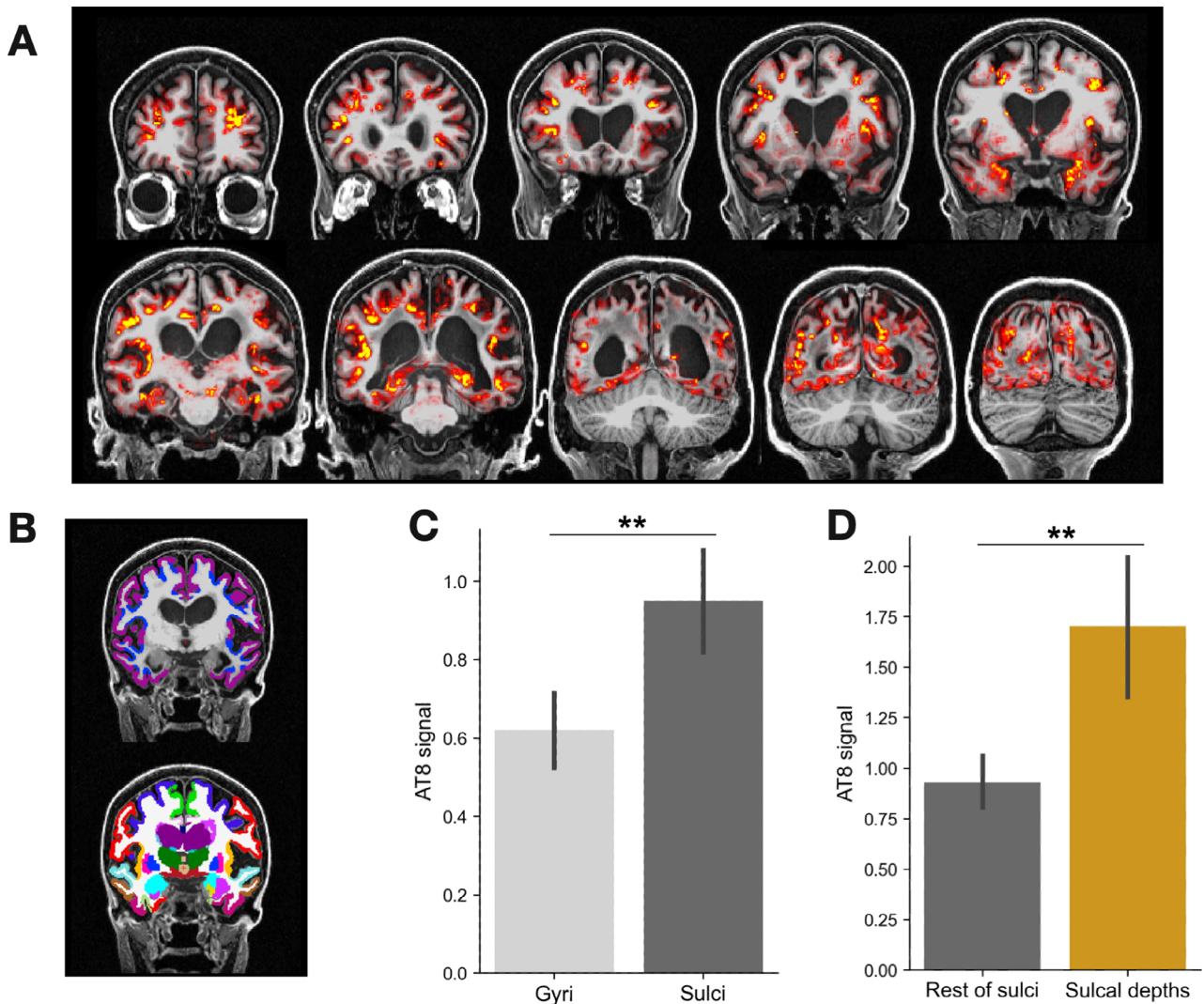


FIGURE 3: Tau accumulates more in sulci in a postmortem Braak 6 case. (A) MRI of a postmortem case with Braak stage 6 tau pathology, with an AT8 (p-Tau Ser202, Thr205 Monoclonal Antibody) histology heat map overlaid. (B) Binarized sulcal (blue) / gyral (purple) masks (top) used to mask DK atlas ROIs (bottom), from which average heatmap signal intensity was calculated. (C) Average AT8 signal intensity across DK atlas cortical regions masked by sulci versus gyri. (D) Average AT8 signal intensity across DK atlas cortical regions further masked by sulcal depths (deepest 5% of sulcal points) versus the rest of sulci (remaining 95%). Error bars = 95% CI. ** = $p < 0.001$. CI = confidence interval; DK = Desikan-Killiany; MRI = magnetic resonance imaging; ROIs = regions of interest.

region of heightened vulnerability to tau pathology, we examined 3T resting state functional connectivity (FC) strength between the hippocampus and the clusters or the rest of the cortex, using a sample of 122 OAs (age = 60–94 years, 53 amyloid positive) and 65 YAs (age = 18–33 years) from the Berkeley Aging Cohort Study (see Supplementary Table S2). We found that hippocampal FC strength was higher in the tau-thickness clusters (Fig 4) than in the rest of the cortex across all groups: a mixed-effects ANOVA revealed a strong main effect of ROI [$F(1, 184) = 42.21, p < 0.0001, \eta^2_p = 0.19$], indicating that FC strength is greater within the clusters. Additionally, there was a significant main effect of group [$F(2, 184) = 7.38, p = 0.0008, \eta^2_p = 0.07$]; post hoc pairwise comparisons

revealed both OA groups had stronger FC strength than the YAs ($A\beta^-$ OA vs YA $p = 0.03$; $A\beta^+$ OA vs YA $p < 0.001$), but that the OA groups were not significantly different ($p = 0.4$). The interaction between ROI and group was not significant [$F(2, 184) = 1.43, p = 0.2409, \eta^2_p = 0.02$], indicating that the observed greater FC strength within the clusters was consistent across groups. These results suggest that the clusters exhibit overall stronger FC to the hippocampus throughout adulthood, and prior to the onset of age-related degeneration or pathology. Whole-brain sulcal versus gyral ROIs did not differ in their hippocampal functional connectivity strength (paired t test $p > 0.4$), indicating that the stronger connectivity observed in the tau-vulnerable clusters cannot be attributed to a global sulcal bias.

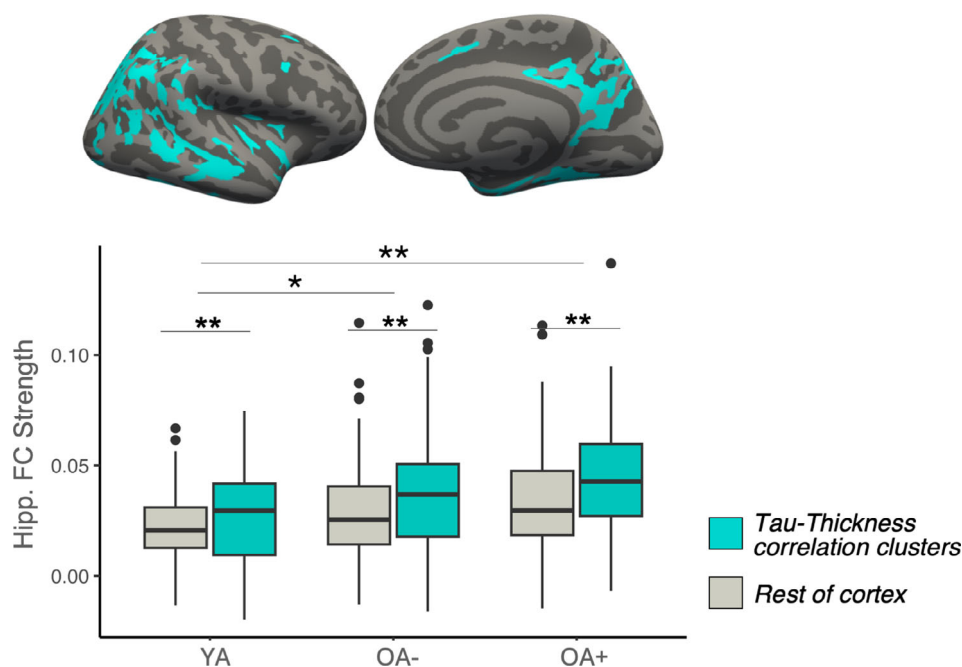


FIGURE 4: Vulnerability of tau-thickness correlation clusters to tau may be related to higher connectivity to the hippocampus. Average hipp. seed-to-voxel resting state FC values in tau-thickness correlation clusters (from Fig 1A) versus the rest of the cortex. The p values report results of mixed-effects ANOVA with fixed effects of group YA/A β -negative OA-/A β -positive OA+ and ROI (clusters/rest of cortex). ** $p < 0.001$, * $p < 0.05$. ANOVA = analysis of variance; FC = functional connectivity; hipp. = hippocampus; OA = older adult; YA = younger adult.

Discussion

In this study, we sought to investigate whether tau pathology primarily affects sulci in AD. We found that tau-related thinning occurs mostly in sulci, and particularly in the depths. Postmortem histological evidence suggests this sulcal vulnerability may occur because sulci accumulate more tau than gyri, and resting state analyses indicate that higher connectivity to the hippocampus throughout the life course in these vulnerable regions may contribute to this phenomenon.

This sulcal vulnerability to tau pathology complements previous findings of sulcal vulnerability to other degenerative processes. For example, atrophy is greater in sulci in aging,⁶ A β preferentially deposits in sulci in AD,^{11,12} and tau pathology has been found to affect sulci in CTE and to cluster around cortical vessels, particularly in depths.^{26,27} Though a much more typical pathological hallmark of CTE, and not a commonly reported characteristic of AD, a slight sulcal bias for tau pathology has also been occasionally reported in AD, although less prominently than in CTE.^{26,27} Taken together with our findings, these observations suggest multiple putative mechanisms underlying sulcal vulnerability. First, prior studies of sulcal-specific atrophy in aging have often overlooked underlying pathology; our results suggest that age-related accumulation of pathological proteins in sulci could partly explain accelerated sulcal thinning with age.

Second, in AD, tau may preferentially localize to sulci through interactions with A β ; further AD-specific pathological processes involving the colocalization of A β and neuritic tau may then occur in sulci, leading to more pronounced atrophy in these regions. Third, vulnerability across AD, aging, and CTE together additionally suggests more generic, non-AD-specific mechanisms at play. Computational modeling studies of traumatic brain injury (TBI) predict that mechanical forces concentrate in sulcal regions, corresponding with tau accumulation patterns observed in CTE.²⁸ This sulcal vulnerability may therefore also relate to repeated mechanical strain in these areas over time, especially given links between AD and CTE: both are mixed 3R/4R tauopathies with similar tau structures,²⁹ and even mild TBIs are associated with increased dementia risk.³⁰

Furthermore, sulcal vulnerability may be influenced by cytoarchitectonics and cerebrovascular dynamics. Studies suggest cell density and vasculature may differ between sulci and gyri,^{31,32} and such differences could plausibly impact connectivity patterns and clearance. Cytoarchitectonic findings are somewhat mixed; whereas the cortical ribbon is well-established to be thicker in gyri, findings on cell density vary, with some reporting greater density in sulci¹⁷ and others reporting no differences between sulci and gyri.³³ Similarly, structural connectivity differences are not fully elucidated, and conclusions differ by methodology.

Although no studies to our knowledge have systematically histologically quantified whether cortical connectivity across the brain biases sulcal or gyral regions (in humans or nonhuman primates), evidence from visual examination of a handful of tracer studies in macaques suggests no clear bias for gyri or sulci.³⁴ With diffusion-based tractography in humans, some studies report lower connection strength and density in sulci versus gyri,³⁵ whereas others (and to our knowledge the only study to investigate connectivity within sulcal depths specifically) report that sulcal depths form a network of long-range core connectivity hubs.¹⁵ These findings, however, are likely influenced by the well-known gyral bias in diffusion tractography, by which gyral connectivity is overestimated compared to true anatomic connectivity due to the orientation of white matter fibers in gyral crowns compared to sulcal fundi.^{34,36} Another factor potentially underlying differences across studies is variation in the definition of sulci and gyri. Some studies define gyri as only the exposed, convex parts of the cortex and sulci as the hidden, concave folds; others use a midpoint along the curvature to divide the cortical ribbon; and some further subdivide the cortex into gyral crests (peaks), gyral walls, sulcal walls, and sulcal depths. In this study, we use the latter 2 approaches, in which we utilize FS's curvature surface to binarize the cortex by sulci and gyri around a curvature-based midpoint (with approximately equal sulcal and gyral area), and also investigate the sulcal depths separately. Future research should further investigate these potential mechanisms underlying selective sulcal vulnerability—including mechanical strain, cerebrovascular dynamics, and structural connectivity patterns—and their relationships with pathology.

The sulcal bias for pathology may both reflect connectivity patterns underlying pathology spread and in turn shed light on the cognitive effects of pathology in these networks. Tau pathology is hypothesized to spread along structural connections, with evidence for transneuronal propagation from cellular, animal, and human studies, and functional connectivity is thought to modulate spread and vulnerability.^{37–40} Given findings that sulcal depths form long-range connectivity hubs,¹⁵ these areas may serve as conduits for an initially sulcal-biased tau spread, leading to heightened vulnerability in AD. In terms of cognitive implications, the hypothesis of fundal cognition suggests that sulcal depths may have a particular role in high-order cognitive processing,¹⁶ and the role of sulcal depths as long-range connectivity hubs could make them critical sites for networks,¹⁵ which, when disrupted, will have particularly severe effects on network function and cognition. Importantly, we note that the observed cognitive and connectivity associations are specific to our predominantly sulcal cluster and are not explained by a general sulcal bias

(in volume or hippocampal connectivity). Thus, the stronger memory associations and hippocampal connectivity we report are specific to the tau-vulnerable clusters identified in our analyses, which are more sulcal. These results suggest that cortical folding alone is not sufficient to explain the observed vulnerability, but rather that sulcal anatomy likely interacts with other factors driving tau-related vulnerability (e.g., cytoarchitecture, structural connectivity) to underlie these regional effects.

Beyond these broad cortical patterns, individual sulci may also exhibit differential vulnerability; for example, the superior temporal sulcus has been found to be particularly vulnerable to AD pathology.^{41,42} Additionally, sulcal anatomy differs between hemispheres, with potential implications for an asymmetric vulnerability to AD-related pathology, as observed here. One putative explanation for our observation that this sulcal-dominant tau-related thinning covers more of the left hemisphere is underlying anatomic differences: though evidence for global hemispheric asymmetries in gyrification are mixed, studies have found a leftward asymmetry in gyrification or surface area in temporal and parietal regions.^{43,44} Additionally, there is some evidence for a leftward asymmetry in the effects of AD.^{45,46} Future studies should further explore how these structural variations contribute to differential disease progression and symptom manifestation at a more anatomically precise level.

It is worth noting that we cannot completely rule out that our results could be influenced by image processing artifacts, including meningeal inclusion, which is known to affect more heavily gyral regions such as entorhinal cortex and to be associated with disease status,^{47,48} potentially introducing additional noise in these regions. However, in our test–retest sample, we showed that gyral thickness does not differ more than sulcal thickness between scans, and that there is no relationship between how sulcal a region is and its thickness test–retest reliability as measured by the ICC between scans. These analyses suggest that gyri are not more prone to measurement noise; however, it is still possible that other systematic biases in segmentation errors exist. Although not introducing biased noise, they could instead result in consistently mis-segmented cortical regions that are not as representative of true biology. Nonetheless, our postmortem sample shows that, both by eye and quantitatively, tau accumulation appears greater in sulci (see Fig 3 and Supplementary Fig S2), adding additional support to our conclusions that is independent of the issue of potential meningeal inclusions.

Despite these findings, several limitations of this study should be considered. Our structural and fMRI analyses were not performed on the same cohort because we

wanted to demonstrate lifelong vulnerability of resting-state networks but did not have access to a cohort that contained MRI, fMRI, and PET data from participants spanning from young adulthood through AD dementia. Additionally, though we infer sulcal vulnerability based on tau accumulation and cortical thickness across the aging and AD spectrum, direct longitudinal evidence linking these processes is lacking. Furthermore, our postmortem validation is limited to a single case because of the difficulty of obtaining these samples and of performing whole-brain postmortem histology and registering to MRI. Future studies using longitudinal imaging and larger post-mortem samples could further clarify the relationships examined here. Future studies could also further investigate the potential mechanisms underlying selective sulcal vulnerability, including cerebrovascular dynamics and structural connectivity patterns, which were outside the scope of this investigation. Nonetheless, here, we provide evidence supporting a sulcal bias for tau pathology and its effects that may both reflect connectivity patterns underlying pathology spread and also shed light on the cognitive effects of pathology in brain networks.

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Author Contributions

S.A.M., K.S.W., and W.J.J. contributed to the conception and design of the study; S.A.M., C.S.F., A.C.M., Y. Chao, Y. Chen, D.U., D.T., and L.T.G. contributed to the acquisition and/or analysis of data; S.A.M., K.S.W., and W.J.J. contributed to drafting the text or preparing the figures.

Potential Conflicts of Interest

W.J.J. has served on data monitoring committees for Lilly, and a Lilly radiotracer was used in this study. The remaining authors declare no competing interests.

Data Availability

All analyses in this study were conducted using software and code from standard neuroimaging and statistical packages, detailed in the Methods section. ADNI data is publicly available at <https://adni.loni.usc.edu/>, and BACS data is available upon request.

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